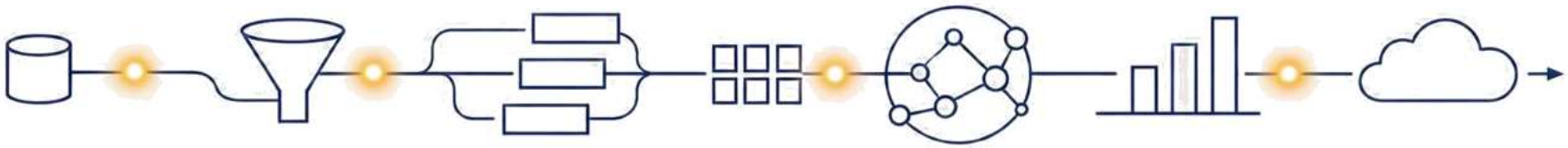
COL2010 · INTRODUCTION TO DATA SCIENCE

Introduction to Data Science

From raw data to real decisions — six modules, one pipeline.



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SEMESTER
Diwali Semester, 2026

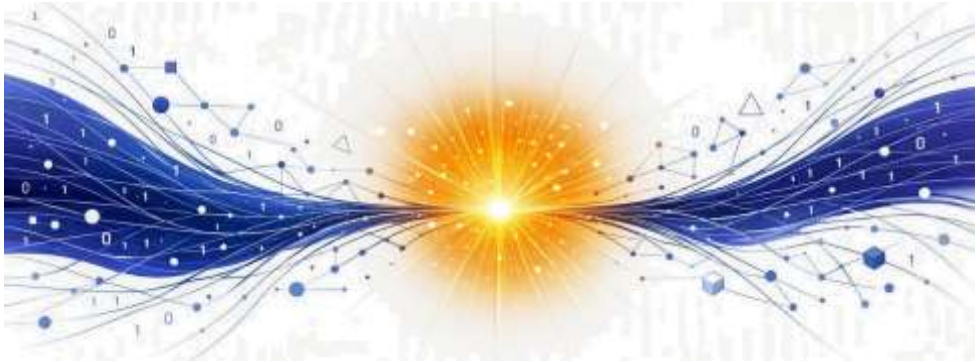
CONTACT
Over Piazza ONLY
https://piazza.com/iitd.ac.in/fall2026/col2010

PART I

The Story of Data Science

01

“What will we learn in this course and why?”



PART I · THE BIG IDEA

What is Data Science?

> **Data Science is the process of converting raw data into useful knowledge.**

THE WORKFLOW



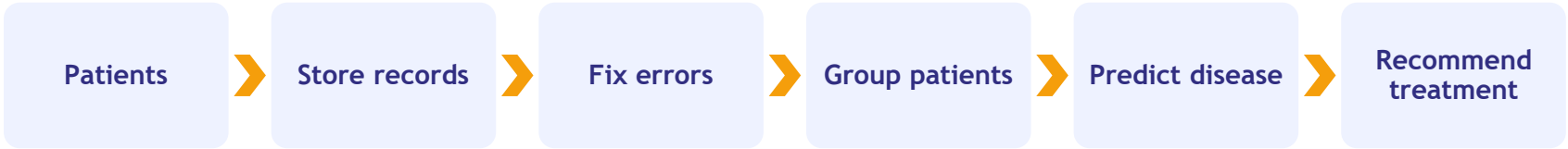
Store
is why the course opens with databases — you can't analyse data you can't reliably keep.

PART I · GROUNDING IT

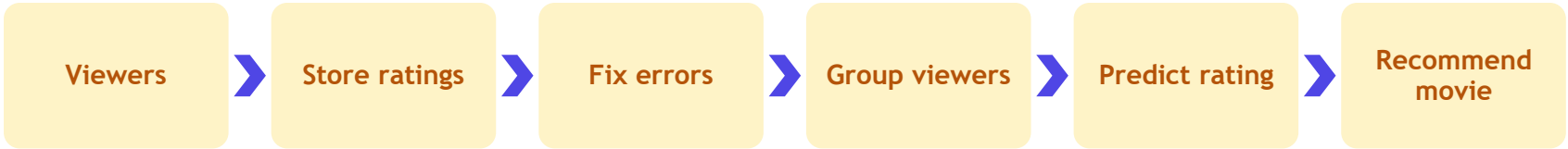
Example

We follow the same journey twice — so every module lands in something concrete.

HOSPITAL DATA



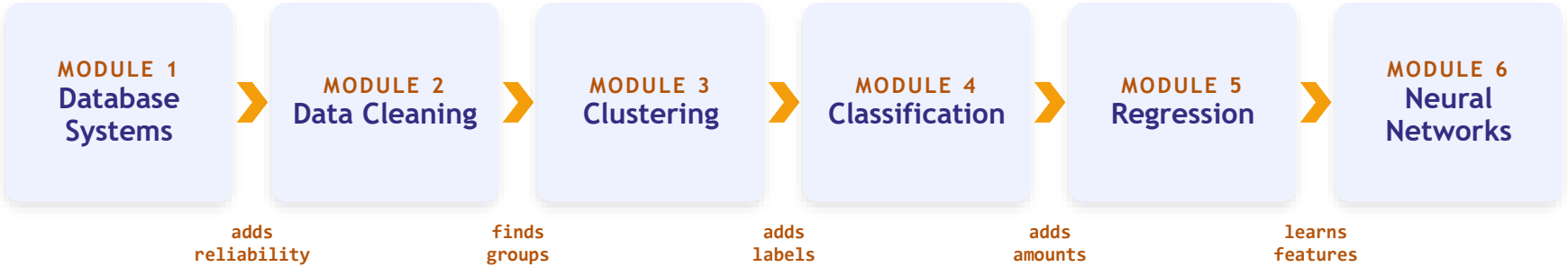
MOVIE RECOMMENDATION



Same six stages, two very different worlds — that is the point.

PART I · THE MAP

The Course Roadmap



The through-line:

every module adds one capability the one before it lacked.

Not six isolated topics — six successive steps in a single data-science workflow.

MODULE 01

Databases

01

THE QUESTION

How do we organize data?

How do we avoid redundancy?

How do we query efficiently?

TOPICS

- ER model
- Relational model
- Normalization
- SQL
- Recursive SQL

WHY THIS MATTERS

Bad storage → bad analysis.

MODULE 02

Data Cleaning

02

THE QUESTION

Can we trust our data?

TOPICS

- Missing values
- Outliers
- Duplicates
- Scaling
- Validation

REMEMBER

Nearly every real ML project spends more time cleaning data than training models.

MODULE 03

Clustering

03

THE QUESTION

What structure exists without labels?

TOPICS

- k-means
- Hierarchical
- Gaussian mixtures
- Spectral clustering

APPLICATIONS

- Customer segmentation
- Image grouping
- Gene analysis

MODULE 04

Classification

04

THE QUESTION

Can we predict categories?

TOPICS

- KNN
- Naive Bayes
- Logistic Regression
- SVM
- Decision Trees
- Random Forests

APPLICATIONS

- Spam detection
- Disease diagnosis
- Fraud detection

MODULE 05

Regression

05

THE QUESTION

Can we predict continuous values?

TOPICS

- Linear regression
- Ridge
- LASSO
- Basis functions
- Regularization

APPLICATIONS

House prices

Demand forecasting

Temperature

MODULE 06

Neural Networks

06

THE QUESTION

Can we automatically learn useful representations?

TOPICS

- Perceptrons
- Hidden layers
- Backpropagation
- Deep learning

APPLICATIONS

Image recognition | Language | Speech | Modern AI

PART I · THE PAYOFF

Putting Everything Together

Read the six modules end to end and the story appears — one continuous pipeline.



Students finally see the story:
raw records become clean data, clean data reveals patterns, and patterns power predictions — for the hospital and the movie site alike.

PART I · PREREQUISITES

Mathematical Background

Stated honestly, here's the toolkit we'll lean on:

✓

Linear Algebra

Vectors, matrices, projections

✓

Probability

Distributions, expectation, Bayes

✓

Basic Calculus

Derivatives, gradients

✓

Programming

Python, for every homework

No prior ML knowledge assumed.

We build every method from these four foundations.

PART I · TOOLING

Software We'll Use

An open-source Python stack — the same tools used in industry and research.

Python
The language of the course

NumPy
Fast numerical arrays

pandas
Tables & data wrangling

matplotlib
Plotting & visualization

scikit-learn
Classic ML algorithms

Jupyter
Interactive notebooks

Full setup instructions will be provided separately.

PART I · PLAN

Reading

PRML — Bishop, Pattern Recognition and Machine Learning
ESL — Hastie, Tibshirani & Friedman, The Elements of Statistical Learning
IDM — Tan, Steinbach & Kumar, Introduction to Data Mining
Database Systems — Garcia-Molina, Ullman & Widom, Database Systems: The Complete Book

Week	Reading
Database Systems	Garcia-Molina: ER model, Relational model, Relational algebra, SQL
Data Cleaning	IDM Chapter 2
Clustering	PRML Chapter 9, PRML Chapter 12; ESL Chapter 14 (selected sections)
Classification	ESL Chapters 2, 9, 12, 15; PRML Chapter 4
Regression	PRML Chapter 3; ESL Chapters 3, 5, 7
Neural Networks	PRML Chapter 5; ESL Chapter 11

What to read for each module of the course.

PART II

Course Administration

02

Everything you need to run the semester — how it works, how you're graded, and how to do well.



PART II · RHYTHM

How the Course Will Run

Your typical week has four moving parts:

1

Lectures

Concepts & intuition — thrice a week

2

Course Forum

Ask questions here — your first stop for help

3

Homework

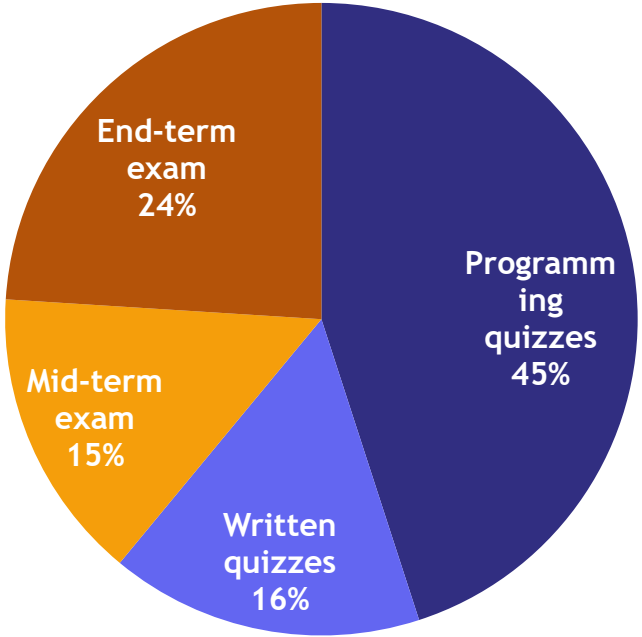
Concepts learnt in class, translate them to programs

4

Office Hours

Scheduled by TAs only when there's demand

Evaluation



- **Programming quizzes** *(best 3-out-of-4)* — 45%
- **Written quizzes** *(best 4-out-of-6)* — 16%
- **Mid-term exam** — 15%
- **End-term exam** — 24%

Passing grade: min. 30% · **A grade:** min. 80% — *absolute vs. relative grading (or a mix) still to be decided*

PART II · HOMEWORK

Homework & Programming Quizzes

1. **Ungraded homework —**
   Module-wise practice sets, not collected or marked
2. **Quizzes come from homework —**
   Programming quizzes are built directly on these problems
3. **Written quiz per module —**
   Ends each module to check you've understood the basic concepts — easy if you pay attention in class
4. **Python only —**
   Using the course's standard stack
5. **Practice pays off —**
   Keep up each week and the quizzes take care of themselves

PART II · GROUND RULES

Academic Integrity

The line is simple: share ideas, never share code.

Use AI to learn how to write programs — but you won't have that crutch in the programming quizzes.

ENCOURAGED

- Discussing concepts at a whiteboard
- Comparing approaches before you code
- Using AI tools to learn — then writing it yourself
- Citing any source or snippet you build on

DISCOURAGED

- Copying a classmate's or online solution
- Sharing your code files with anyone
- Pasting AI-generated code
- Paraphrasing code

Consequences:

Any ethics violation — plagiarism, attendance proxies, or cheating on quizzes — means zero on the work and referral to the disciplinary committee.

PART II · EXPECTATIONS

Attendance

75% required

Institute policy — attend at least 75% of lectures, or your grade is reduced.

No recordings

Lectures aren't recorded, but slides are uploaded after class.

Sick or emergency leave

Counts within your 25% — there's no separate medical exemption.

Missed quizzes

No make-up quizzes — the best-3-of-4 rule already covers a miss.

PART II · STAYING IN TOUCH

Communication

Office hours
First stop for real help

Discussion forum
Post here — everyone benefits

Email
For personal or private matters

Response time:
expect a reply within 48 working hours.

BEFORE YOU EMAIL, CHECK:

- Is it answered in the FAQ?
- Is it in an LMS announcement?
- Is it a course-wide question?

If yes, use the forum instead.

PART II · ADVICE

How to Succeed

1. **Attend lectures**
   Momentum compounds week over week.

2. **Read before class**
   Arrive with questions, not blank pages.

3. **Start homeworks early**
   Debugging always takes longer than you think.

4. **Write code yourself**
   Typing it is how it sticks.

5. **Ask questions**
   Every good one helps ten classmates too.

6. **Understand, don't memorize**
   Concepts transfer; formulas fade.

AND FINALLY

One Question

How can a computer learn from data?

That is the question this entire course attempts to answer.

